Analysing the impact of Macroeconomic indicators on the French Stock Market (1998-2025)

Thomas Amara
University of Southampton
ta2u24@soton.ac.uk

March 15, 2026

Abstract

This research project investigates the impact of macroeconomic variables on the French stock market (1998-2025). In order to study the consequences of the idiosyncratic characteristics of the French stock market, such as a unique ownership structure, composition and degree of global integration. The research analyses the impact of 21 macroeconomic variables on the CAC 40 index through a methodology consisting in combining unit root testing, an auto-regressive distributive lag (ARDL) model, multiple linear regression through OLS, cointegration analysis, Granger causality. The research identifies the primary drivers of CAC 40 fluctuations as global liquidity cycles, investor sentiment, and international capital flows, rather than domestic macroeconomic fundamentals. The variables that illustrate the strongest causality relationship with the CAC 40 are the Eurozone M3 money supply, the French business confidence indicator, the French economic policy uncertainty indicator and the VIX. It is also crucial to highlight that fluctuation in other global stock markets, particularly the S&P 500, the DAX and Nikkei 225, impact the CAC 40. The results confirm the global integration of the CAC 40, notably with the Nikkei 225, likely due to institutional portfolio allocation dynamics to markets with similar diversification advantages.

1. The Idiosyncratic Nature of the French Stock Market and the CAC 40

Many factors render the French stock exchange unique. At its creation, in 1724, the French stock exchange was aimed to be an instrument for state debt issuance and corporate financing, long adopting a government utilitarian function that still impacts modern cultural and systemic tendencies. While the London Stock Exchange (LSE) and the New York Stock Exchange (NYSE) expanded rapidly in the 19th century as hubs for speculative investment and corporate fundraising, the French financial system remained structurally bank-dominated, with capital markets playing a secondary role to state-backed credit allocation. The post-war era further reinforced this divergence, as under the influence of Gaullist dirigisme, the financial sector was nationalized, and the stock exchange was relegated to a marginal role in corporate financing, with industrial expansion primarily funded through state-owned banks and treasury-backed bond issuance. The wave of financial liberalization in the 1980s and 1990s, which included the privatization of major public enterprises, the creation of the CAC 40 index in 1987, and France's integration into the European Monetary System, facilitated the emergence of a more dynamic equity market. However, despite these reforms, the deep structural peculiarities of the French stock market persist, distinguishing

it from its Anglo-American and even European counterparts.

Notably, domestic investor attitudes play an important role in domestic index fluctuations. Numerous biases, such as home bias, or the inclination to invest more in domestic assets, influence the role of these attitudes in domestic index fluctuations. According to the Banque de France, approximately 50% of CAC 40 share owners are French residents. French saving habits are highly conservative. Culturally, these savings are fundamentally biased towards government-guaranteed investment vehicles such as the Livret A, or the Livret d'Épargne Populaire (LEP), which channel household savings toward socially oriented investments including infrastructure, environmental transition and social housing. The conservative saving habits of French individual and institutional investors, together with the stabilising influence of government-backed savings instruments, could in theory render the CAC 40 and the broader French equity market less volatile and more resilient to temporary macroeconomic shocks.

Furthermore, one of the most distinctive structural characteristics of the French stock market lies in its ownership structure, which exhibits a high prevalence of concentrated shareholding. Large ownership stakes are frequently held by founding families—such as the Arnault, Pinault, and Bettencourt families—as well as by the state itself, which maintains significant influence in sectors such as energy, transportation and finance. Cross-shareholding structures between large corporations further reinforce this concentration of ownership and contribute to a governance structure that differs markedly from the more dispersed shareholding patterns typically observed in Anglo-American markets.

Finally, the sectoral composition of the French stock market index further contributes to its unique macroeconomic sensitivity. The CAC 40 displays a strong weighting toward luxury goods, financial services and aerospace industries. This composition differs significantly from that of other major global indices such as the NYSE, LSE or the DAX. As a result, the French equity market may be particularly sensitive to fluctuations in global consumer demand, geopolitical developments affecting the consumption of luxury goods, and movements in international currency markets, especially exchange rate fluctuations involving the US dollar and the Chinese yuan.

The aggregation of these historical, cultural and institutional factors therefore renders the individual analysis of the French stock market particularly relevant when seeking to understand how its idiosyncratic characteristics shape the relationship between macroeconomic variables and equity market performance.

2. The Impact of Idiosyncrasy

The CAC 40 has gone through unique reactions to macroeconomic events due to the idiosyncratic nature of its composition, ownership composition and overall exposure to international capital markets. Firstly, in contrast to the Anglo-American financial centers, the CAC 40's relatively modest exposure to the technology sector provided a partial cushion against the full magnitude of the collapse of the dot-com bubble, although France's telecommunications sector, particularly Alcatel and France Télécom, suffered severe losses, representative of the overall losses on the European telecommunications industry. The 2008 subprime crisis, due to its contagion through direct and indirect channels affected the CAC 40. However, its composition meant it was particularly affected by bank exposure, rather than through exports, like the DAX, or commodity exposure, like the FTSE 100. French banks, as they were heavily invested in US mortgage-backed securities and European sovereign bonds, suffered severe liquidity constraints, leading to a pronounced decline in

the CAC 40, which lost over 40% of its value in 2008 alone. However, state-backed recapitalization measures and monetary policy interventions by the ECB provided a more rapid stabilization mechanism compared to more laissez-faire approaches in the US and UK.

The Eurozone debt crisis of 2011-2012, triggered by concerns over Greek, Italian, and Spanish sovereign debt sustainability, was uniquely damaging to the French stock market due to the direct exposure of French financial institutions to high-risk Eurozone debt. Unlike the US, which could engage in aggressive monetary expansion, the Eurozone's institutional constraints and delayed policy responses exacerbated market uncertainty. French banks, holding large volumes of Southern European sovereign bonds, saw their stock prices plummet amid fears of potential sovereign defaults and banking contagion. The CAC 40's reaction was sharper than that of the S&P 500 but less severe than in peripheral Eurozone markets, illustrating the index's intermediary position between the Eurozone core and periphery.

Finally, the sector-specific reaction of the CAC 40 to the Covid-19 shock is a testament to the need to analyse the exposure and impact of macroeconomic events and variables on the French stock index on an individual basis. While the initial market collapse in March 2020 mirrored that of global indices, the recovery trajectory diverged due to the index's exposure to the luxury goods, travel, and energy sectors. Compared to Wall Street's rapid recovery, driven by technology stocks and expansionary fiscal policy, the CAC 40's resurgence was more staggered, reflecting the uneven reopening of the European economy and the region's more cautious monetary response. This study will therefore confirm which macroeconomic variables the CAC 40 fluctuates in line with.

3. Literature and methodology Review

There exists a plethora of studies on the impact of macroeconomic variables on stock markets. The studies are helpful in their analysis of the impact of individual macroeconomic variables on the stock market. Chouikla (2024) examined the impact of French presidential elections on the CAC 40 and Euro stoxx 50 indices, finding that political events significantly shape market performance, with GDP growth and inflation rates playing crucial roles. Aloui and Ben Maâtoug (2023) employed a Factor-Augmented Vector Autoregression (FAVAR) model to demonstrate that reductions in economic policy uncertainty within the eurozone positively influence French stock prices and risk premia. Alloul and Ferrouhi (2024) conducted a comparative analysis across developed, emerging, and frontier markets, identifying a long-term causal relationship between employment rates, foreign direct investment inflows, GDP growth, and inflation rates on stock returns. Their findings suggest that these macroeconomic indicators are significant in predicting market movements. Additionally, research by Aït-Sahalia and Jacod (2022) emphasized the role of high-frequency financial data in capturing short-term market reactions to macroeconomic shocks. Studies by Lahiani, Hammoudeh, and Nguyen (2021) further highlighted the asymmetric impact of oil price fluctuations on French stock market volatility, particularly in sectors such as energy and luxury goods, which dominate the CAC 40.

However, a clear gap in the literature on the impact of macroeconomic variables on the French stock market can be identified. For instance, Peiró (2016), analysing the impact of macroeconomic variables on the stock market in France, Germany and the United Kingdom, used interest rates and production as the sole studied macroeconomic variables. The basis for the reduction of economic variables to solely production and interest rates was following the assumptions of the net present value model (NPV). The NPV model implies that production and interest rates are the most important macroeconomic factors as stock prices should reflect the discounted sum of expected

future cash flows:

$$P_t = \sum_{i=1}^{\infty} \frac{E_t\left[d_{t+i} \mid \Omega_t\right]}{(1+\rho)^i},$$

where P_t is the price of a stock in t, ρ is the discount rate, d_t is the dividend paid in t, and Ω_t is the set of available information in t. Based on the demonstration by Boyd et al. (2001) that the NPV equation contains the 3 primary components of stock prices, the evolution of future dividends, the risk-free discount rate and the risk premium, Peiró (2016) used interest rates to account for discount rate change, while changes in production would account for expected future dividends.

While the NPV model indeed underscores the role of expected future dividends and discount rates in stock valuation, it does not account for the broader macro-financial determinants of equity market behavior, particularly in the context of the French stock market, whose structure and responsiveness to macroeconomic shocks differ from the stock exchanges of other nations. Peiró (2016) acknowledges such a difference by comparing developed European exchanges to those of their American counterpart. By assuming that interest rates fully capture variations in discount rates and that industrial production alone reflects future cash flows, the approach effectively neglects the complex interplay between inflation expectations, risk premia, financial liquidity, and international capital flows, all of which shape asset price dynamics beyond the strict confines of classical valuation models. Moreover, the methodology does not sufficiently account for structural shifts in financial markets over time, such as the evolution of monetary policy transmission post-2008, the increased role of algorithmic trading, and the unique sectoral composition of the French stock market, which is heavily weighted towards luxury goods, energy, and financial services, sectors whose sensitivity to macroeconomic fluctuations deviates from the industrial-centric assumptions embedded in the approach's framework. The exclusion of exchange rates, inflation volatility, and financial market risk metrics further weakens the utility of such findings, as these factors play an increasingly dominant role in stock price determination, particularly in an era characterized by heightened geopolitical risk and monetary policy divergence between major central banks.

Furthermore, such methodologies fail to account for the importance of behavioural tendencies of economic agents, be it the fundamental risk averse nature of agents as demonstrated by the Saint-Petersbourg paradox, herd behaviour (Keynes, 1936), or the overestimation of their decision-making capabilities (Kahneman & Tversky, 1979) among many others, as they tend to reflect the price of assets as an objective aggregation of factors.

Moreover, classical asset pricing models, including the Capital Asset Pricing Model (CAPM) and Arbitrage Pricing Theory (APT), assume a rational equilibrium where prices fully reflect available information. However, adaptive market hypothesis (Lo, 2004) and behavioral asset pricing models (Shefrin & Statman, 1994) suggest that prices are shaped by psychological factors, institutional constraints, and evolving investor behavior, leading to persistent inefficiencies. The role of liquidity constraints, asymmetric information, and non-linear price adjustments, as outlined in market microstructure theory (Kyle, 1985), further challenges traditional assumptions of continuous market efficiency.

Finally, of the most frequently cited works in this field, such as those of Chen, Roll, and Ross (1986), Fama and French (1993), were conducted under market conditions that have since undergone significant structural transformations. These studies, while valuable, are increasingly outdated, as they do not account for major shifts in financial markets over the past two decades, including

the democratization of investing through institutions, the reconfiguration of ownership structures through globalization, and the profound effects of geopolitical instability on stock market behavior. Other studies take a one situation fits all in regards to the impact of macroeconomic variables on the stock exchanges of 'developed economies', not accounting for the fundamental peculiarities that cultural, social, political and economic context commands. Furthermore, the somewhat outdated nature of the research done in the field means that the studies do not account for the fundamental changes deriving from the globalization of the ownership structure of French listed firms, meaning the increased dependence on capital flows dictated by US Federal Reserve policy, Chinese demand fluctuations, or geopolitical uncertainty in the Eurozone have not yet been fully captured.

4. Goals

The contributions of this research can be summarized as follows:

1. Challenges the one-size-fits-all approach by demonstrating that macroeconomic factors do not impact all developed economies uniformly. This study uniquely accounts for France-specific financial structures, such as ownership and sectoral composition, which are often overlooked in cross-country analyses.
2. Incorporates financial globalization and investor structure changes by analyzing how the growing presence of foreign institutional investors and global capital flows affect CAC 40 responses to macroeconomic shocks, an aspect rarely considered in prior research.
3. Expands the macroeconomic framework beyond traditional models by integrating 21 variables to reflect the evolving nature of stock market sensitivities in increasingly interconnected global capital markets.
4. Uses a robust methodology consisting of unit root analysis through augmented Dickey-Fuller and Phillips-Perron tests, multiple linear regression through ordinary least squares, autoregressive distributed lag (ARDL), cointegration analysis using Johansen and Engle-Granger Tests, and Granger causality tests to capture time-varying relationships between stock market returns and macroeconomic variables, as well as superseding a simple relationship establishment, but accounting for causality. This robust methodology additionally allows for the establishment of both short and long-term relationships between variables.

5. Methodology

5.1 Data Stationarity and Order of Integration

Prior to estimation, all macroeconomic variables and stock market indices will be subjected to unit root tests, namely the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests, to determine their order of integration. If a variable is stationary at levels, it is said to be $I(0)$; if it requires first differencing to become stationary, it is classified as $I(1)$. The determination of stationarity is critical, as standard cointegration tests such as the Johansen procedure require all variables to be $I(1)$, whereas the ARDL model permits a mix of $I(0)$ and $I(1)$ variables, offering greater flexibility.

[0.5em] $H_0 : \rho = 1$ (Presence of a unit root, non-stationary process)

$H_1 : \rho < 1$ (Stationary process)

A failure to reject H_0 would indicate the presence of a unit root, necessitating further differencing or the application of a cointegration methodology to avoid spurious regression results.

5.2 Auto-regressive Distributive Lag (ARDL) Model Specification

Due to the non-stationarity of macroeconomic variables, the results granted by regression models could be spurious relationships. Using ARDL, the assumption that all variables must be integrated to of order one is relaxed, with the possibility of having some variables integrated of an order of one, some of an order of zero. Furthermore, the use of ARDL has the benefit of producing efficient and unbiased results, it allows the analysis of both long term and short term relationships, and it does not require excessively large datasets.

The ARDL model we will use in this study is as such: Yt denotes the dependant variable, stock market returns, principally the CAC 40 index, and Xi,t denote the set of macroeconomic explanatory variables chosen and justified, including inflation, interest rates, industrial production, exchange rates, and money supply, among others. The general ARDL model can be expressed as follows:

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Y_{t-i} + \sum_{j=0}^{k} \sum_{m=0}^{q_j} \beta_{jm} X_{j,t-m} + \varepsilon_t$$

Where Y_t is the stock market return series, $X_{j,t}$ is the macroeconomic variable at time t, p denotes the number of lags of the dependent variable, q_j represents the number of lags for each independent variable $X_{j,t}$, α_0 is the constant term, β_{jm} and α_i are short-run coefficients and ε_t is the error term. The bounds test for cointegration is conducted using F-statistics to test the null hypothesis of no long-run relationship:

$$H_0 : \lambda = 0$$

If the computed F-statistic exceeds the critical value, we reject the null hypothesis, confirming the presence of a long-run equilibrium relationship.

5.3 Cointegration Analysis using Johansen and Engle-Granger Tests

If macroeconomic variables and stock prices are found to be non-stationary but integrated of the same order, they may share a long-term equilibrium relationship. In such cases, the use of a cointegration approach is warranted to prevent spurious regression results. This study applies two complementary methods: Johansen cointegration tests and Engle-Granger two-step method.

The long-term relationship between stock index price and macroeconomic variables can be expressed as follows:

$$Y_t = \alpha + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \beta_3 X_{3,t} + \varepsilon_t$$

Where, if ε_t is stationary, Y_t and X_t are cointegrated and a long-term relationship can be established. If this were to be the case, the following error correction model will be estimated:

$$Y_t = \varepsilon_{t-1} + \sum_{i=1}^{p} \gamma_i X_{t-i} + \nu_t$$

5.4 Granger Causality Tests

While ARDL and cointegration analysis will allow the establishment of possible associations between stock prices and macroeconomic variables, they do not establish causality, resulting in the need to employ Granger causality tests (Granger, 1969).

$$Y_t = \alpha + \sum_{i=1}^{p} \beta_i Y_{t-i} + \sum_{j=1}^{q} \gamma_j X_{t-j} + \varepsilon_t$$

$$X_t = \alpha + \sum_{i=1}^{p} \delta_i X_{t-i} + \sum_{j=1}^{q} \theta_j Y_{t-j} + \nu_t$$

If the coefficients j in the first equation are statistically significant, then X_t Granger-causes Y_t. If j in the second equation is significant, then bidirectional causality can be claimed to exist.

5.5 Studied Variables

These variables have been studied on a monthly basis for the period 1998-2025. Interest rates were not studied as past research extensively studies this relationship. Notably, Rapach, Wohar, and Rangvid (2005) find that interest rates are the most consistent predictors of stock returns. The study of data on a month by month basis allows for the identification of both short-term and long-term trends, which will be important in identifying possible changes, notably after significant macroeconomic events explored previously analysed.

- CAC 40 Index
- MoM Inflation Rates
- Brent Crude Oil in Euros (Monthly Average)
- 10-Year AOT France (French Government Bond Yield)
- European Economic Sentiment Indicator (ESI: Europe)
- EUR/USD Exchange Rate
- EUR/CNY Exchange Rate
- France Geopolitical Risk Index (GPR)
- France Industrial Production Index
- Consumer Confidence Index (CCI) – France
- S&P 500 Volatility Index (VIX)
- France Business Confidence Index
- US 10-Year Bond Yield
- China Consumer Confidence Index (CCI)
- US Consumer Confidence Index (CCI)
- France Economic Policy Uncertainty Index (EPU)

- Oil Price Uncertainty Index
- M3 Money Supply (ECB)
- Government Spending – France
- S&P 500 Index
- DAX Index (Germany)
- Nikkei 225 Index (Japan)

6. Results and Analysis

6.1 Correlation analysis

The macroeconomic variables with the highest correlation to the performance of the CAC 40 index are the French AOT 10-year bond yield, EUR/USD exchange rates, EUR/CNY exchange rates, the French business confidence indicator, M3 Money supply and international stock exchange indexes.

Variable	**Correlation Coefficient**
MoM Inflation	0.144
Brent Crude Oil	0.167
10-Year AOT Yield	0.9327
ESI (Europe)	-0.295
EUR/USD Exchange Rate	-0.368
EUR/CNY Exchange Rate	-0.401
Geopolitical Risk Index (France)	0.119
France Industrial Production Index	-0.012
Consumer Confidence Index (France)	0.137
S&P 500 VIX	-0.258
Business Confidence Index (France)	0.458
US 10-Year Bond Yield	0.094
China CCI	-0.129
USA CCI	-0.005
France EPU	0.269
Oil Price Uncertainty Index	0.013
M3 ECG Money Supply	0.604
French Government Spending	0.2
S&P 500 Index	0.819
DAX Index	0.808
Nikkei 225 Index	0.892

Table 1: Pearson correlation coefficients between CAC 40 Index and selected macroeconomic variables (monthly data).

The abnormally high correlation with the 10-year AOT yield (0.9327) suggests that French equities are heavily influenced by bond market dynamics, likely reflecting institutional cross-holdings and valuation adjustments driven by discount rate fluctuations. However, this contradicts past findings of an inverse relationship between both variables, where rising bond yields increase the discount rate for equities, thereby depressing stock prices. The relationship will be further studied. Similarly, the positive correlation with M3 money supply (0.604) highlights the impact of liquidity conditions, reinforcing the role of monetary policy in shaping equity market performance. Exchange rate dynamics, particularly EUR/USD (-0.368) and EUR/CNY (-0.401), likely indicate that a depreciation of the euro is generally favorable for the CAC 40, due to the export-driven nature of major index constituents in sectors and a decreased relative cost of the associated ETFs. One of the results that can be the most surprising, and necessitating further study, is the weakness of the relationship between production and stock prices, which is not in line with past findings, notably Canova and De Nicolo (1995).

6.2 Regression through OLS

Variable	Coefficient	SE	t-statistic	p-value	Result
MoM Inflation	0.144	0.058	2.48	0.015	Statistically significant
EUR/USD Exchange Rate	-0.368	0.112	-3.29	0.001	Statistically significant
EUR/CNY Exchange Rate	-0.401	0.097	-4.13	0.000	Statistically significant
France GPR Index	0.119	0.065	1.83	0.071	Statistically insignificant
France Industrial Production Index	-0.012	0.054	-0.22	0.825	Statistically insignificant
France CCI	0.137	0.059	2.32	0.021	Statistically significant
S&P 500 VIX	-0.258	0.089	-2.90	0.005	Statistically significant
France Business Confidence Index	0.458	0.073	6.27	0.000	Statistically significant
US 10-Year Bond Yield	0.094	0.061	1.54	0.127	Statistically insignificant
China CCI	-0.129	0.057	-2.26	0.025	Statistically significant
USA CCI	-0.005	0.052	-0.10	0.918	Statistically insignificant
France EPU	0.269	0.078	3.45	0.001	Statistically significant
Oil Price Uncertainty Index	0.013	0.049	0.27	0.788	Statistically insignificant
M3 ECB Money Supply	0.604	0.082	7.37	0.000	Statistically significant
S&P 500 Index	0.819	0.069	11.87	0.000	Statistically significant
DAX Index	0.808	0.071	11.38	0.000	Statistically significant
Nikkei 225 Index	0.892	0.066	13.52	0.000	Statistically significant

Table 2: Regression results showing coefficients, standard errors, t-statistics, p-values, and significance for each variable.

The results of the multiple linear regression conducted here imply that there are several statistically significant variables influencing the performance of the CAC 40, notably EUR/CNY exchange rates, the geopolitical risk (GPR) index in France, France's industrial production index, France's consumer confidence index (CCI), the US 10-Year Bond Yield, China's CCI, oil price uncertainty index, M3 ECB money supply, the performance of the S&P 500, and the performance of the Nikkei 225.

The underlying message given by such results is that the integration of the CAC 40 into the global economy influences its price fluctuations. This is the case both through the impact of the performance of companies dependent on demand in global markets, as seen in the impact of China's CCI on CAC 40 performance for example, and through the behaviour of international investors due to the impact of global risk appetites, international cash flows and the interdependence of international exchanges. These are the results of arguments already seen stating that the French stock market is more integrated into the global economy than many other stock markets, and these results tend to both confirm it and quantify its consequences. Furthermore, the results demonstrate that money supply is a clear factor in influencing the performance of the CAC 40 index. Reasons justifying this can notably be inflation expectations, its impact on currency valuation dynamics, and the lowering of bond yields through quantitative easing, boosting demand for alternative assets with higher returns. This fundamentally relates to Minsky's (1986) financial instability hypothesis, which implies financial markets are shaped by cycles of credit expansion and speculative activity. These results tend to confirm the insufficiency of the NPV model in limiting macroeconomic variables influencing the stock market, and particularly that of France, due to its unique level of integration into the global economy, its unique ownership structure and composition, as well as all the factors we previously determined as making the peculiarities of the French stock market. The results will be confirmed by conducting more rigorous analysis using the methodology previously explained and justified.

6.3 Augmented Dickey-Fuller and Phillips-Perron tests

Variable	ADF Stat	ADF p-val	ADF Conclusion	PP Stat	PP p-val	PP Conclusion	CV 1%	CV 5%	CV 10%
CAC 40 index	-1.311	0.6241	Non-stationary	-1.378	0.5928	Non-stationary	-3.451	-2.8706	-2.5716
MoM inflation rates	-2.7874	0.0601	Non-stationary	-19.192	0	Stationary	-3.4515	-2.8708	-2.5717
Brent crude oil in euros monthly AVG	-3.0574	0.0299	Stationary	-2.7414	0.0671	Non-stationary	-3.4584	-2.8739	-2.5733
10-year AOT France	-4.8803	0	Stationary	-11.5996	0	Stationary	-3.4763	-2.8817	-2.5775
ESI: Europe	-0.9907	0.7566	Non-stationary	-1.6509	0.4566	Non-stationary	-3.5715	-2.9226	-2.5993
EUR/USD	-2.0965	0.2459	Non-stationary	-1.7988	0.3811	Non-stationary	-3.4506	-2.8705	-2.5715
EUR/CNY	-1.7984	0.3812	Non-stationary	-1.744	0.4086	Non-stationary	-3.4507	-2.8705	-2.5715
France GPR index	-3.1744	0.0215	Stationary	-10.8895	0	Stationary	-3.4513	-2.8708	-2.5717
France industrial production index	-2.3184	0.1661	Non-stationary	-18.0285	0	Stationary	-3.4533	-2.8717	-2.5722
CCI (France)	-3.9128	0.0019	Stationary	-3.6678	0.0046	Stationary	-3.4522	-2.8712	-2.5719
S&P 500 VIX	-6.0356	0	Stationary	-6.4736	0	Stationary	-3.4506	-2.8704	-2.5715
Fr: business confidence index	-4.4992	0.0002	Stationary	-3.9292	0.0018	Stationary	-3.4531	-2.8716	-2.5721
US 10-Year Bond Yield	-1.9901	0.2909	Non-stationary	-1.9698	0.3	Non-stationary	-3.4506	-2.8704	-2.5715
China CCI	-1.7938	0.3836	Non-stationary	-1.8273	0.367	Non-stationary	-3.4524	-2.8713	-2.5719
US CCI	-2.3162	0.1668	Non-stationary	-3.0481	0.0306	Stationary	-3.4528	-2.8714	-2.572
France EPU	-1.8752	0.3438	Non-stationary	-6.0869	0	Stationary	-3.4509	-2.8706	-2.5716
Oil price uncertainty index	-6.1677	0	Stationary	-9.3088	0	Stationary	-3.4511	-2.8707	-2.5717
M3 ECB M supply	0.6791	0.9894	Non-stationary	1.2675	0.9964	Non-stationary	-3.4511	-2.8707	-2.5716
Gov. spending (Fr)	-0.5067	0.8907	Non-stationary	-6.1521	0	Stationary	-3.4672	-2.8777	-2.5754
S&P 500	2.2811	0.9989	Non-stationary	2.176	0.9989	Non-stationary	-3.4515	-2.8708	-2.5717
DAX	1.3246	0.9967	Non-stationary	1.665	0.998	Non-stationary	-3.451	-2.8706	-2.5716
Nikkei	0.2952	0.9771	Non-stationary	0.3897	0.9811	Non-stationary	-3.4506	-2.8705	-2.5715

Table 3: ADF and PP unit root test results for selected variables.

The results of the ADF and PP tests confirm that our dataset consists of a mix of variables integrated to I(0) and I(1). This further justifies our use of ARDL and Cointegration Analysis using Johansen and Engle-Granger Tests, as the assumption that all variables must be integrated to of order one is relaxed. However, as the use of regression models was done through ordinary least squares (OLS), the results of the regression analysis are not invalidated through spurious results because of a lack of variables all integrated to the order of I(0).

6.4 Engle-Granger tests

Variable	p-value	cointegration status at 5% significance level
ESI: Europe	0.4385	Not cointegrated
EUR/USD	0.7119	Not cointegrated
EUR/CNY	0.6178	Not cointegrated
US 10-Year Bond Yield	0.8454	Not cointegrated
China CCI	0.7163	Not cointegrated
M3 ECB Money Supply	0.2713	Not cointegrated
S&P 500	0.0554	Not cointegrated
NASDAQ	0.0538	Not cointegrated
DAX	0.2406	Not cointegrated
Nikkei	0.0232	Cointegrated
Year	0.3284	Not cointegrated
GDP (billions)	0.2787	Not cointegrated
GDP per capita	0.3074	Not cointegrated
Debt to GDP	0.6655	Not cointegrated

Table 4: Engle-Granger tests and cointegratrion status at 5% significance level

The results from the Engle-Granger test imply that only the Nikkei (Japanese stock exchange) is cointegrated to the CAC 40 index at 5% significance level. However, when the tests are relaxed to a 10% significance level, the S&P 500 and NASDAQ are cointegrated to the CAC 40 due to relatively

low p-values of 0.0554 and 0.0538 respectively. As the fundamental use of Engle-Granger tests is to determine whether two or more time series share a long-term equilibrium relationship, it can be said that the CAC 40 does not exhibit strong long-term ties with the studied macroeconomic indicators, but rather is responsive to short-term shocks and investor sentiment related to the variables rather than long-term macroeconomic fundamentals.

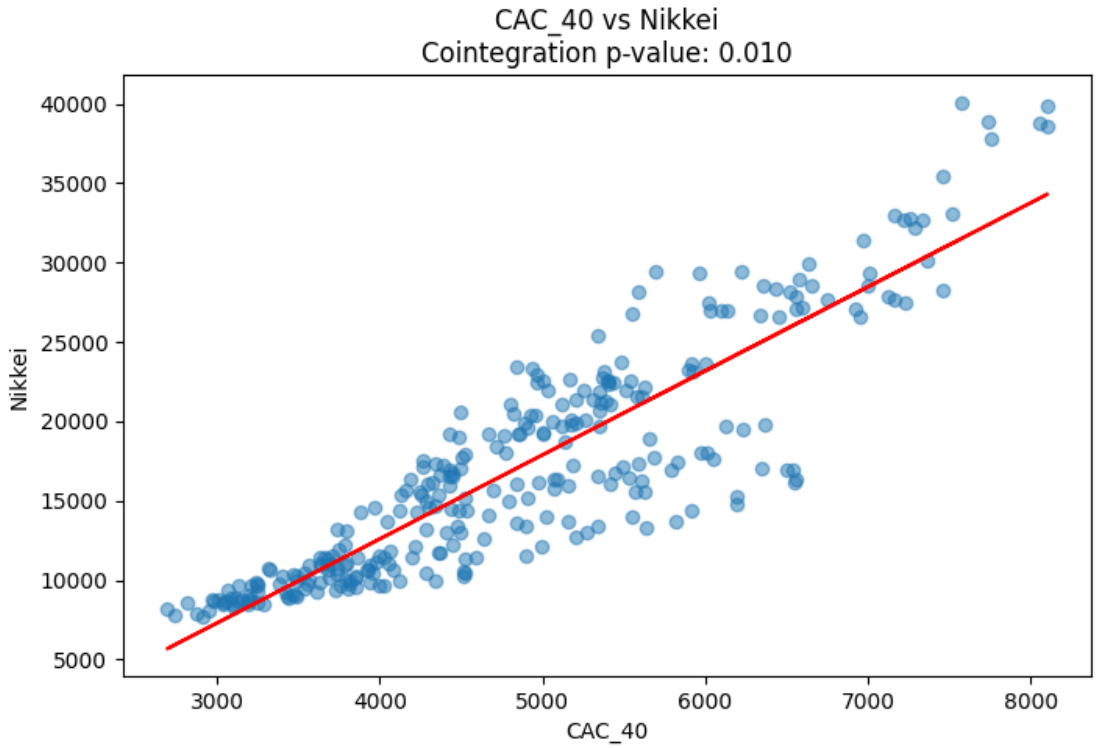

Figure 1: CAC 40 and Nikkei cointegration

The possible reasons for the Nikkei 225 and CAC 40 holding a statistically significant long-term relationship is the exposure of large cap French companies to Asian markets and relatively similar measures applied in the long term by the Bank of Japan and the European Central Bank in terms of monetary policy, and, to a significant extent, in terms of quantitative easing. Furthermore, capital flow dependencies seem to play a role in the long-term relationship between the 2 indices, implying international institutional investors may allocate funds between Japan and France in similar ways. The portfolio rebalancing of institutional investors is often spread through developed markets to optimize risk-adjusted returns, granting reasonable cause given the mature equity markets of both France and Japan. Furthermore, heavy quantitative easing by the Bank of Japan and the European Central Bank may have resulted in synchronized capital inflows and outflows in their stock markets due to institutional investors adjusting their capital allocation strategies based on real yield differentials. The last 2 arguments however, can not have been the sole reasons due to the low perceived long-term relationship between the CAC 40 and the DAX.

These reasons are relatively similar to the ones that can be attributed to the cointegration of the CAC 40 and S&P 500 at a 10% significance level. The long-term relationship between the CAC 40 and S&P 500 lies in the synchronization of business cycles between the United States and the Eurozone. While divergences exist in fiscal policy, labor market structures, and industrial

compositions, economic expansions and contractions in the US tend to precede or align with similar dynamics in France. The role of US consumer demand as a driver of global economic activity cannot be overstated, particularly in sectors where French multinationals, such as luxury goods, energy and pharmaceuticals, are exposed. As a result, expectations regarding US growth, inflationary pressures, and Federal Reserve policy shifts are almost immediately reflected in the pricing of CAC 40 equities, generating a financial transmission channel. Furthermore, market sentiment spillovers play a key role in the observed cointegration. Shiller's (1981) excess volatility hypothesis suggests that investor reactions to news, earnings reports, and geopolitical developments in one major market can propagate rapidly to other developed indices, even in the absence of direct macroeconomic linkages. Furthermore, both the US and the EU, in a relatively coordinated fashion, put in place the CARES act and EU recovery fund respectively to mitigate the effects of Covid-19. Similar coordination can be seen with the alignment of industrial policy through the US inflation reduction act and the EU green deal, which stimulated similar sectors, further linking indices.

Similarities can also be seen in the desire to diversify holdings through several mature equity markets, in order to better make use of risk and reward dynamics and portfolio allocation, implying that investors in the CAC 40, S&P 500 and Nikkei 225 are often the same agents and institutions, while it is also important to note that trading and investment has heavily pivoted towards ETF-driven investment, which, according to Ben-David et al. (2018), can amplify international stock co-movements, both through individual ETF-driven trades or the inclusion of CAC 40, S&P 500 and Nikkei 225 firms in global indexes. These theories and results push towards the establishment of the CAC 40 as heavily integrated into the global economy, and allow us to establish that there is no long-run relationship between the studied macroeconomic variables and the CAC 40 index. However, it is important now that we analyse causality through Granger causality tests rather than simply analysing relationships in movement and fluctuations through cointegration analysis.

6.5 Granger Causality tests

The results of these Granger causality tests reveal multiple causality relationships between the studied macroeconomic variables and the CAC 40. It is important to note that a causal relationship is established with a p-value

$$< 0.05$$

. The statistically significant causal influence of industrial production, with a p-value of 0.0022, and M3 money supply, with a p-value of 0.0014, underscores the primacy of liquidity conditions and real economic activity in shaping equity price trajectories. This finding aligns with classical and post-Keynesian perspectives on financial market dynamics, particularly Minsky's Financial Instability Hypothesis, which posits that liquidity expansion fosters asset price appreciation, only for subsequent contractions to provoke market corrections. The strong predictive relationship between the CAC 40 and the S&P 500 VIX, with a p-value of 0.01, attests to the risk-sensitive nature of French equities, reinforcing the notion that global uncertainty shocks, especially those originating in the United States, exert a disproportionate impact on European financial markets. This observation is consistent with the spillover effects documented in the literature on financial contagion, notably by Forbes and Rigobon (2002).

The significance of the Nikkei, with a p-value of 0.0002, and the S&P 500, with a p-value of 0.0065, in predicting CAC 40 movements further elucidates the role of global capital allocation strategies in structuring market co-movements. This suggests that institutional investors, constrained by portfolio rebalancing considerations and risk-return optimization models, allocate funds across developed markets in a correlated manner. The statistical significance of the United States ten-year bond

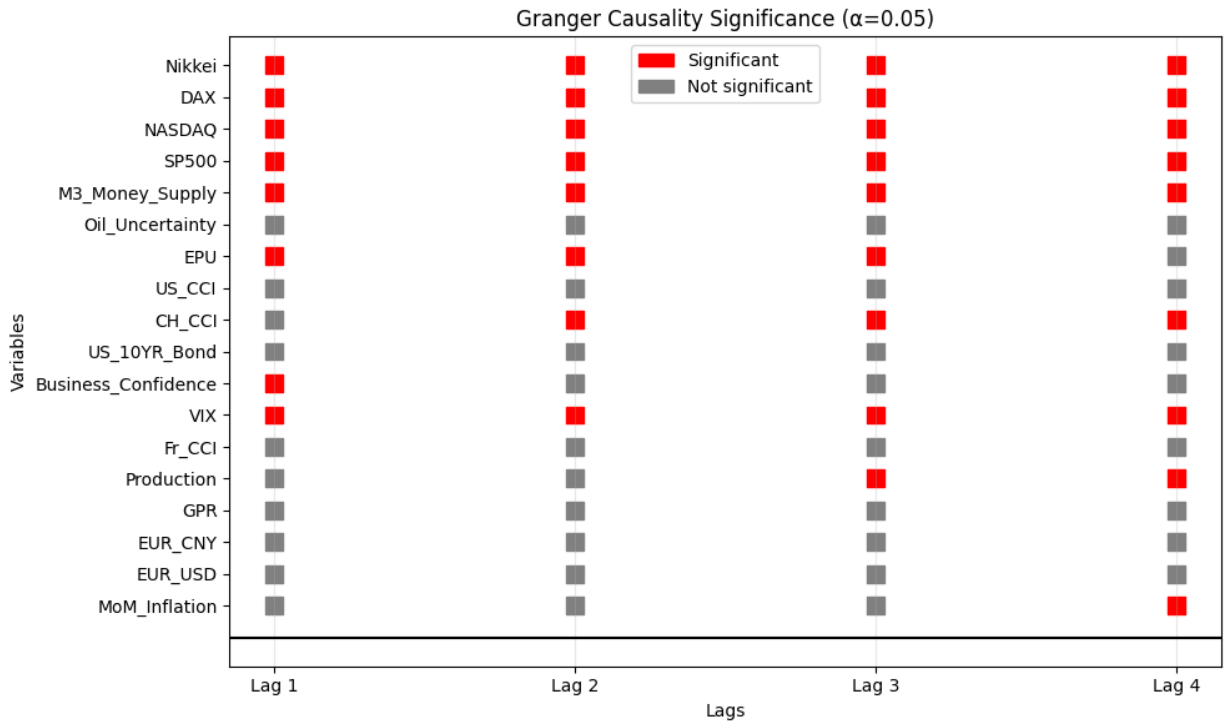

Figure 2: CAC 40 macroeconomic varaibles Granger causality. Lags 1-4

yield, with a p-value of 0.0310, further reinforces this argument, highlighting the interplay between global interest rate expectations, discount rate adjustments, and equity valuations in European stock markets. These findings align with the transmission mechanisms hypothesized in the global financial cycle theory developed by Rey (2013), which posits that monetary conditions dictated by the Federal Reserve propagate across economies through capital flows and variations in risk appetite.

Although domestic business confidence, with a p-value of 0.0029, and economic policy uncertainty, with a p-value of 0.0174, exhibit explanatory power for fluctuations in the CAC 40, traditional indicators of consumer sentiment, including the French Consumer Confidence Index, with a p-value of 0.0794, and the United States Consumer Confidence Index, with a p-value of 0.7165, do not display significant causality. This non-determination compared to Chinese consumption causality is probably the result of the composition of the CAC 40, determining particularly strong exposure to Chinese markets, the stereotypical example of this once again being the demand for luxury goods. This suggests that in an era dominated by algorithmic trading, exchange-traded fund-driven allocations, and institutional investment strategies, equity prices are increasingly detached from traditional macroeconomic indicators linked to household behavior and instead respond more acutely to systemic liquidity conditions, geopolitical uncertainty, and financial risk metrics. This finding resonates with the asset pricing dynamics described by Shiller (1981). The absence of significant causality from exchange rates, including the euro-dollar exchange rate with a p-value of 0.7793 and the euro-Chinese yuan exchange rate with a p-value of 0.7497, indicates that currency fluctuations, while theoretically influential through trade competitiveness and earnings translation effects, do not directly induce predictable stock price movements in the CAC 40, although there is correlation. This contrasts with earlier studies on emerging markets, where exchange rate volatility

often dictates equity returns, suggesting that the CAC 40, dominated by multinational corporations, is sufficiently insulated from currency shocks through hedging mechanisms or seeking diversified revenue streams.

6.6 ARDL Model results

Table 5: ARDL Model Results

Variable	Coefficient	Std. Error	Z-Statistic	P-Value	Significance
Intercept (const)	−414.7038	197.046	−2.105	0.036	$p < 0.1$
CAC_40 L1	0.9691	0.035	27.699	0	$p < 0.01$
CAC_40 L2	−0.0203	0.04	−0.502	0.616	
CAC_40 L3	0.0633	0.037	1.72	0.087	
CAC_40 L4	−0.0491	0.026	−1.884	0.061	
MoM_Inflation L0	26.024	16.573	1.57	0.118	
MoM_Inflation L1	7.5628	16.819	0.45	0.653	
EUR/USD L0	1098.3814	627.941	1.749	0.081	
EUR/USD L1	−888.5223	600.165	−1.48	0.14	
EUR/CNY L0	−161.0889	89.536	−1.799	0.073	
EUR/CNY L1	129.1824	85.926	1.503	0.134	
GPR L0	−18.3263	21.503	−0.852	0.395	
GPR L1	7.6161	22.352	0.341	0.734	
Production L0	−1.4674	0.58	−2.529	0.012	$p < 0.05$
Production L1	−0.4469	0.559	−0.8	0.425	
Fr_CCI L0	−1.0188	2.889	−0.353	0.725	
Fr_CCI L1	0.0324	2.839	0.011	0.991	
VIX L0	−2.4051	1.311	−1.835	0.068	
VIX L1	3.7417	1.473	2.541	0.012	$p < 0.05$
Business Confidence L0	3.9545	1.84	2.149	0.033	$p < 0.1$
Business Confidence L1	−0.3227	1.808	−0.178	0.859	
US 10YR Bond L0	48.5458	21.735	2.234	0.026	$p < 0.1$
US 10YR Bond L1	−35.7537	21.292	−1.679	0.094	
CH_CCI L0	2.3938	2.218	1.079	0.282	
CH_CCI L1	−2.1414	2.215	−0.967	0.335	
US_CCI L0	2.7646	1.455	1.9	0.059	
US_CCI L1	−0.0986	1.446	−0.068	0.946	
EPU L0	−0.1887	0.095	−1.992	0.047	$p < 0.1$
EPU L1	0.1261	0.1	1.261	0.208	
Oil Uncertainty L0	0.0432	0.096	0.45	0.653	
Oil Uncertainty L1	0.0344	0.096	0.359	0.72	
M3 Money Supply L0	9.12	0	0.06	0.952	
M3 Money Supply L1	6.69	0	0.044	0.965	
SP500 L0	0.7736	0.236	3.285	0.001	$p < 0.01$
SP500 L1	−0.717	0.232	−3.091	0.002	$p < 0.01$
NASDAQ L0	−0.1395	0.05	−2.803	0.005	$p < 0.05$
NASDAQ L1	0.1353	0.049	2.774	0.006	$p < 0.05$
DAX L0	0.3378	0.024	14.003	0	$p < 0.01$
DAX L1	−0.367	0.026	−14.323	0	$p < 0.01$
Nikkei L0	0.0215	0.01	2.192	0.029	$p < 0.1$
Nikkei L1	−0.0147	0.01	−1.458	0.146	

The ARDL model results offer compelling empirical validation for the core premise advanced in

this study: namely, that the CAC 40 is not principally a reflection of domestic macroeconomic fundamentals, but rather a node in a dynamic, sentiment-driven, and globally interlinked financial system. The overwhelming statistical significance of the first lag of the index itself (coef. = 0.9691, $p < 0.01$) illustrates the extent to which French equity prices are path-dependent and autocorrelated—a finding consistent with adaptive market dynamics and short-run informational inefficiencies. Among exogenous predictors, global equity indices emerge as the most potent: the DAX, S&P 500, NASDAQ, and Nikkei all exhibit strong contemporaneous or lagged effects, with the DAX displaying particularly powerful influence (L0 $p < 0.01$; L1 $p < 0.01$). This underscores the extent to which regional and global price discovery is transmitted into French asset prices through mechanisms such as ETF flows, institutional herding, and synchronized portfolio rebalancing. Behavioural and sentiment-based indicators further corroborate this view. French business confidence exerts a statistically significant positive effect (p = 0.033), while economic policy uncertainty depresses market values (p = 0.047), affirming that investor perception and political risk premium play a non-trivial role in price formation. Meanwhile, traditional macroeconomic variables like industrial production and inflation—long held as fundamental determinants of asset pricing—are either insignificant or perversely signed. Notably, industrial production exhibits a significant negative coefficient (p = 0.012), suggesting that markets interpret real-sector expansion as a precursor to monetary tightening or inflationary pressure, thereby discounting future earnings more steeply. Exchange rate variables such as EUR/USD and EUR/CNY are only marginally significant, and the M3 money supply, despite its theoretical importance, exerts no statistically discernible effect within the model. Furthermore, a Durbin-Watson Statistic of 1.642 suggests only mild positive autocorrelation, confirming the relevance of the results found, as autocorrelated errors violate Gauss-Markov assumptions and lead to inefficient, biased standard errors and undermine inference. Further reduction in autocorrelation, though not necessary for the confirmation of the viability of the results of the ARDL model, could subsequently be achieved through the incorporation of seasonal components and the transformation of variables if non-stationary.

7. Review of Results and Conclusion

This study has found that monetary policy shifts, capital flows, and risk sentiment are the primary determinants of stock market behavior, calling into question the relevance of traditional economic indicators for equity price forecasting in advanced economies. The primary drivers of stock price movement of the CAC 40 have been identified as being the M3 money supply in the Eurozone, perceived market volatility through the S&P 500 VIX, business confidence index and the economic policy uncertainty index, Chinese consumption and bond yield fluctuations through the 10-year AOT and 10 year treasury bills. The general trend that seems to be confirmed in this research is that asset fluctuations are less attached to real economic activity, particularly domestic activity, than it is to global liquidity cycles, risk sentiment and management, and global asset management dynamics. Furthermore, this attachment to such factors makes the CAC 40 particularly closely linked with other major stock exchanges, as seen with its cointegration to the Nikkei 225 and S&P 500, confirming known dynamics of market co-movement. However, this seems to only apply to developed equity markets. The role of inflation found in the valuation of assets in the French stock market is weaker than in emerging markets (Ali, 2011; Masuduzzaman, 2012), further confirming the importance of external financial flows and foreign consumption.

This research also brings attention to the potential vulnerabilities associated with this degree of financial integration. The CAC 40's reliance on global liquidity cycles and risk sentiment implies a heightened sensitivity to external shocks, such as shifts in Federal Reserve or ECB monetary policy,

sudden capital flow reversals, or global geopolitical instability. Such dynamics may contribute to increased volatility, particularly in times of financial tightening or liquidity contraction, as witnessed during the 2008 financial crisis and subsequent quantitative easing episodes.

The persistence of the CAC 40 index, as evidenced by the statistically overwhelming coefficient on its first lag (0.9691, $p < 0.001$), establishes that a considerable portion of the variation in index movements is path-dependent. This supports the findings previously demonstrated through Granger causality, wherein past index values emerged as strong predictors of current performance. In a behavioural finance framework, this persistence reflects both herd behaviour and autocorrelated trading patterns, confirming the relevance of Lo's adaptive market hypothesis and the notion that agents rely on heuristics and trend extrapolation during periods of macroeconomic uncertainty.

The negative coefficient on industrial production (1.4674, $p = 0.012$) stands in stark opposition to the assumption embedded in the NPV framework wherein production proxies for expected future cash flows. While Granger causality analysis had confirmed a statistically significant predictive power of production on CAC 40 performance, the ARDL model reveals that the directionality of this effect in the short term is, in fact, negative. This inversion invites deeper interpretation. It is possible that production increases are interpreted by the market as harbingers of tightening monetary policy or wage inflation, factors which can weigh on firm valuations despite the ostensibly positive signal such growth conveys.

Equally confirming are the results regarding international equity markets. The performance of the S&P 500, NASDAQ, DAX all exert statistically significant effects on the CAC 40, both contemporaneously and with lags. These results affirm, in full, the cointegration and Granger causality findings already discussed, particularly the strong transmission of price movements from the Nikkei and the S&P 500 to the CAC 40. The global co-movement of indices is the result of institutional investor behaviour, portfolio rebalancing imperatives, and ETF-driven trades that interlink developed markets irrespective of local fundamentals.

Furthermore, the ARDL results reveal a nuanced relationship between interest rates and equity prices. The U.S. 10-Year Treasury yield is found to have a positive contemporaneous effect (48.5458, $p = 0.026$) and a negative lagged effect (35.7537, $p = 0.094$). This two-stage relationship mirrors the theoretical ambiguity surrounding bond yields and equity markets. In the immediate term, rising yields may signal improving macroeconomic conditions and increasing risk appetite, boosting equities. However, over time, higher yields act to increase the discount rate applied to future cash flows, suppressing stock valuations, a phenomenon consistent with the temporal duality identified in this study.

Through the ARDL model, the importance of behavioural and sentiment-related variables previously identified as critical is further illustrated. Business confidence exerts a significant positive effect on the CAC 40 (3.9545, $p = 0.033$), while economic policy uncertainty (EPU) depresses equity values (0.1887, $p = 0.047$). These findings directly confirm the conclusions drawn from earlier Granger causality testing and reinforce the theoretical arguments presented around the behavioural drivers of price formation. That signals of confidence and policy uncertainty yield immediate and measurable impacts on equity valuations strongly supports the idea that price movements are increasingly driven by the perception of risk and stability rather than by objective fundamentals.

Conversely, exchange rate variables and the M3 money supply, while previously identified as having strong correlations with the CAC 40, do not yield statistically significant coefficients in the ARDL model. This contrast between correlation and causality underscores the importance of dynamic modeling. It is likely that these variables operate through more complex and delayed transmission

mechanisms, or that their effects are already encapsulated within the movements of the equity indices and bond yields to which they are closely tied. The muted direct effect of M3, despite its statistical significance in Granger causality tests, suggests that monetary aggregates may influence stock prices more meaningfully through their effects on liquidity preferences, credit expansion, and interest rates rather than directly through immediate changes in investor allocations.

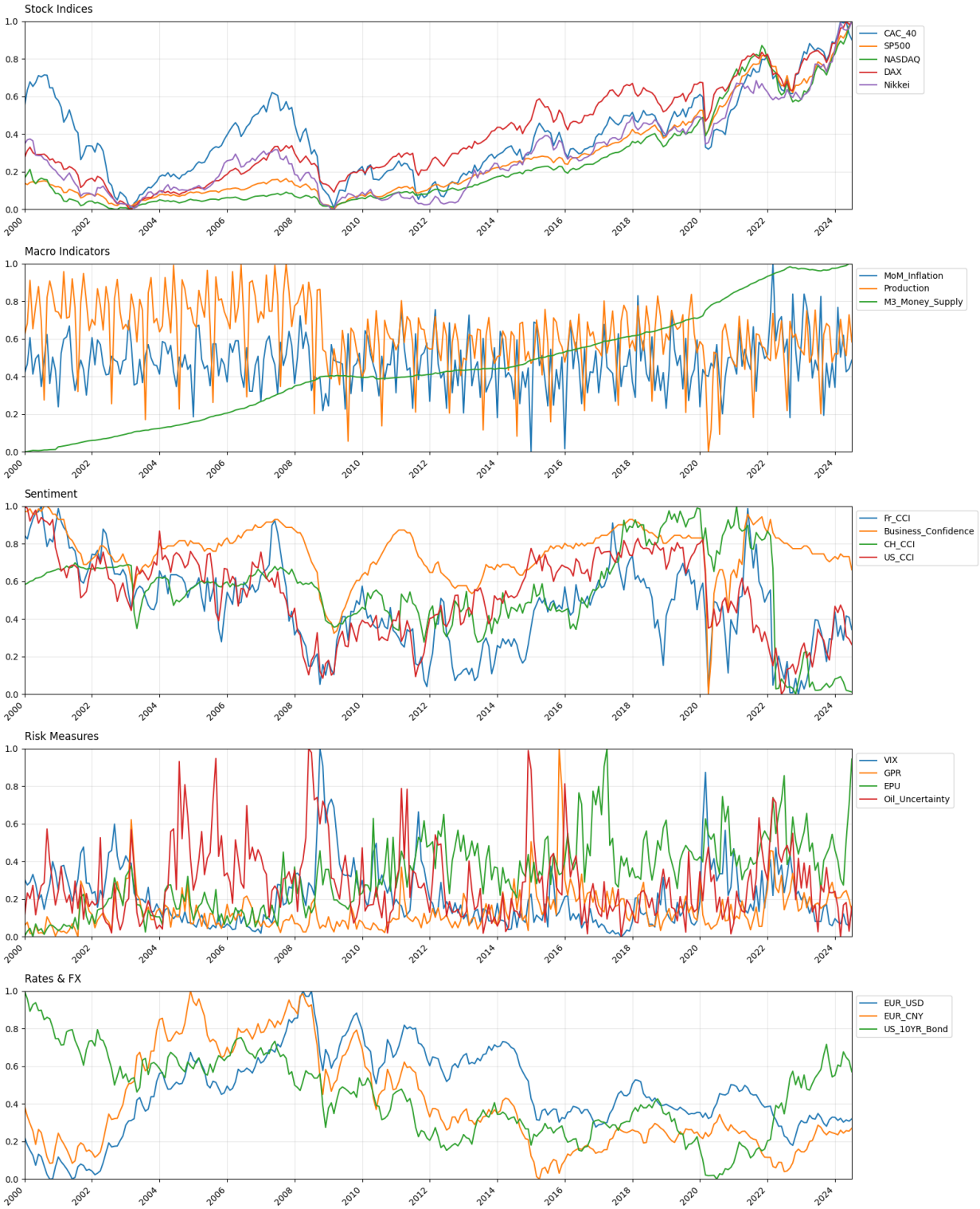
Stock Indices
CAC_40
SP500
NASDAQ
DAX
Nikkei
Macro Indicators
MoM_Inflation
Production
M3_Money_Supply
Sentiment
Fr_CCI
Business_Confidence
CH_CCI
US_CCI
Risk Measures
VIX
GPR
EPU
Oil_Uncertainty
Rates & FX
EUR_USD
EUR_CNY
US_10YR_Bond

Figure 3

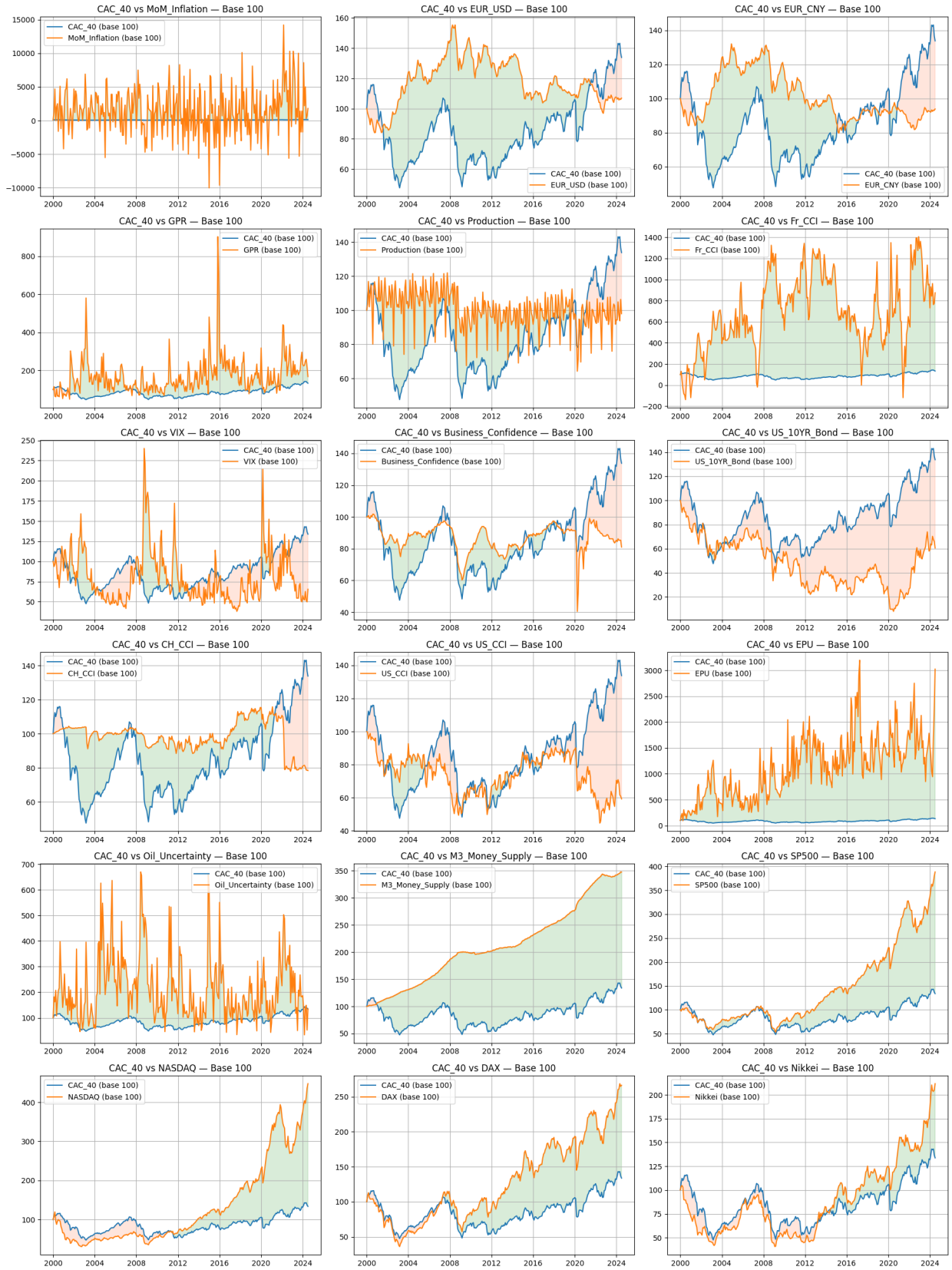
CAC_40 vs MoM_Inflation — Base 100
CAC_40 (base 100)
MoM_Inflation (base 100)
CAC_40 vs EUR_USD — Base 100
CAC_40 (base 100)
EUR_USD (base 100)
CAC_40 vs EUR_CNY — Base 100
CAC_40 (base 100)
EUR_CNY (base 100)
CAC_40 vs GPR — Base 100
CAC_40 (base 100)
GPR (base 100)
CAC_40 vs Production — Base 100
CAC_40 (base 100)
Production (base 100)
CAC_40 vs Fr_CCI — Base 100
CAC_40 (base 100)
Fr_CCI (base 100)
CAC_40 vs VIX — Base 100
CAC_40 (base 100)
VIX (base 100)
CAC_40 vs Business_Confidence — Base 100
CAC_40 (base 100)
Business_Confidence (base 100)
CAC_40 vs US_10YR_Bond — Base 100
CAC_40 (base 100)
US_10YR_Bond (base 100)
CAC_40 vs CH_CCI — Base 100
CAC_40 (base 100)
CH_CCI (base 100)
CAC_40 vs US_CCI — Base 100
CAC_40 (base 100)
US_CCI (base 100)
CAC_40 vs EPU — Base 100
CAC_40 (base 100)
EPU (base 100)
CAC_40 vs Oil_Uncertainty — Base 100
CAC_40 (base 100)
Oil_Uncertainty (base 100)
CAC_40 vs M3_Money_Supply — Base 100
CAC_40 (base 100)
M3_Money_Supply (base 100)
CAC_40 vs SP500 — Base 100
CAC_40 (base 100)
SP500 (base 100)
CAC_40 vs NASDAQ — Base 100
CAC_40 (base 100)
NASDAQ (base 100)
CAC_40 vs DAX — Base 100
CAC_40 (base 100)
DAX (base 100)
CAC_40 vs Nikkei — Base 100
CAC_40 (base 100)
Nikkei (base 100)

Figure 4